

AHU FDD / RCx (single-system)

Offline pandas-oracle pack for one AHU folder or a building folder of device subfolders. This is not the BUILDING_100 Overview-mirrored lab PDF.

AHU_1

1. Data health

Role coverage and physical bounds. Out-of-range counts are a quality pre-screen, not confirmed faults.

role	coverage %	min	max	out of range	bound
discharge-air-temp	100.0	55.0	55.0	0	30..150
mixed-air-temp	100.0	57.0	81.8	0	-20..110
return-air-temp	100.0	75.0	75.0	0	40..100
outside-air-temp	100.0	35.0	200.0	1	-60..130
duct-static-pressure	100.0	0.2	0.2	0	-1..8
duct-static-pressure-sp	100.0	1.2	1.2	0	
fan-cmd	100.0	1.0	1.0	0	
fan-status	100.0	1.0	1.0	0	
outside-air-damper	100.0	0.0	0.0	0	

2. Sensor validation

Pandas oracle SV rules when the mapped sensors exist. Skipped rows are missing roles, not passes.

rule	status	fault_h	missing roles
SV-RANGE	FAULT	1.0	
SV-FLATLINE	FAULT	71.0	
SV-SPIKE	FAULT	2.0	
SV-STALE	PASS	0.0	

3. Anomaly screening

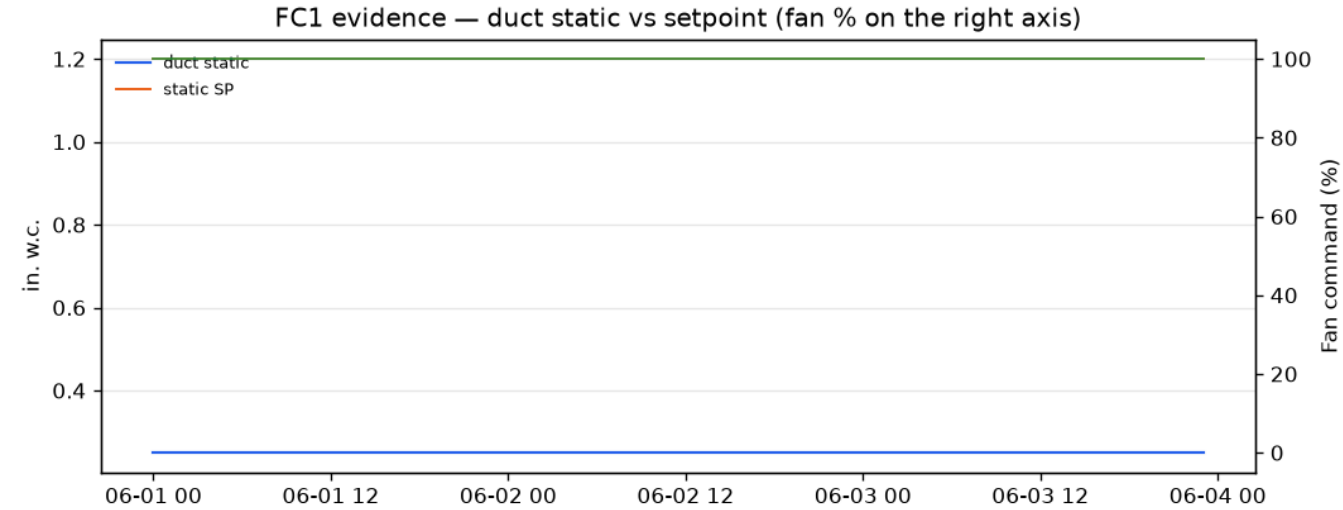
Unsupervised anomaly minutes are not FDD faults. Z-score and MAD are SQL-portable screens; STL and Isolation Forest are Python-only. Cookbook FC1 and economizer rules apply a fan-ON operational gate. Do not read the scoreboard as confirmed faults. Mapped fan is ON for 100.0% of samples.

point	role	method	anomaly min	events	skipped
DAT	discharge-air-temp	nan	0.0	0	binary_like
MAT	mixed-air-temp	zscore	0.0	0	
MAT	mixed-air-temp	mad	0.0	0	
RAT	return-air-temp	nan	0.0	0	binary_like
OAT	outside-air-temp	zscore	0.0	0	
OAT	outside-air-temp	mad	0.0	0	
DSP	duct-static-pressure	nan	0.0	0	binary_like
DSPSP	duct-static-pressure-sp	nan	0.0	0	binary_like
SF_C	fan-cmd	nan	0.0	0	binary_like
SF_S	fan-status	nan	0.0	0	binary_like
OAD	outside-air-damper	nan	0.0	0	binary_like

4. Duct static / FC1

Supply-fan duct-static fault (cookbook FC1): static below setpoint while the fan is at high speed.

rule	status	fault_h	missing roles
FC1	FAULT	72.0	



5. Economizer performance

Mixing and economizer cookbook rules (FC2, FC3, FC10, FC11, ECON-1, ECON-2, ECON-4).

rule	status	fault_h	missing roles
FC2	PASS	0.0	
FC3	PASS	0.0	
FC10	SKIPPED_MISSING_ROLES		cooling-valve
FC11	SKIPPED_MISSING_ROLES		discharge-air-temp-sp, cooling-valve
ECON-1	FAULT	26.0	
ECON-2	PASS	0.0	
ECON-4	PASS	0.0	

6. Economizer scatter

economizer_delta_scatter: $x = \text{OAT} - \text{RAT} (\text{delta_or_f})$, $y = \text{MAT} - \text{RAT} (\text{delta_mr_f})$. Reference lines $y = \text{OA fraction} * x$ for 0/25/50/75/100% OA. Fan ON and $|\text{OAT}-\text{RAT}| \geq 10$ F. Do not plot OAT-MAT vs RAT-MAT.

Economizer free-cooling delta scatter (fan on, $|OAT - RAT| \geq 10^\circ F$)

